

Jacob Sanders

Forward Deployed / Applied AI Engineer

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SUMMARY

Forward-deployed / applied AI engineer and two-time founder who ships production systems end-to-end — from board bring-up to cloud-native SaaS — over 13+ years. Thrives in ambiguous, high-ownership environments: scopes vague problems, ships working software in weeks rather than quarters, and iterates directly with users. Rare breadth-plus-depth profile — full-stack (React, Next.js, TypeScript, serverless AWS) built on deep systems foundations (bare-metal C/C++, BLE, distributed blockchain infrastructure) — with the domain-ramp speed to go from unfamiliar problem to shipped solution across wildly different verticals. AI-native builder: ships production LLM applications on the Claude API — multi-agent pipelines, structured extraction, semantic search, and context-grounded assistants — with guardrails, model-tiered cost control, and leak-free evaluation harnesses.

EXPERIENCE

Lien Into Revenue Services, LLC

Co-Founder

Huntington Beach, CA (Remote)

August 2025 – Present

- Co-founded and developed a production SaaS platform that manages the full lifecycle of personal injury chiropractic cases, from patient intake through lien negotiation and settlement collection
- Built the frontend in Next.js (App Router) and React, delivering a responsive role-based UI for chiropractors and case managers
- Designed the serverless AWS backend integrating DynamoDB, Lambda, AppSync (GraphQL), Cognito, S3, and CloudFront via AWS Amplify
- Implemented lien negotiation tooling enabling users to track outstanding liens, manage settlement offers, and monitor collection status across active cases

ARK Electronics

Full-Stack Engineer (1099 Contract)

Remote

February 2025 – April 2025

- Embedded with the client to scope requirements from scratch and ship a cloud-based manufacturing traceability platform end-to-end in weeks — covering product setup, serial number generation, barcode artifact creation, and test-result tracking for production and QA workflows
- Built a React/TypeScript/Vite (CoreUI) frontend and a fully serverless AWS backend using Lambda, AppSync (GraphQL), API Gateway (REST), DynamoDB, S3, and Cognito
- Automated serial generation with collision checks and batch writes; generated QR-code PDFs stored in S3; built test result ingestion and file upload endpoints with a legacy data migration path
- Exposed dual API surfaces (GraphQL for the web UI, REST for external clients) with API key management, usage plans, throttling, and event-driven job status updates for long-running generation batches

The Honeydew App, Inc.

Founder

Salt Lake City, UT (Remote)

April 2023 – February 2025

- Founded an on-demand handyman services marketplace; owned all system architecture and the majority of engineering while managing 4 University of Utah CS interns across frontend and backend workstreams
- Built the consumer-facing Flutter mobile application using an MVVM architecture backed by a serverless TypeScript/Supabase backend covering authentication, real-time job dispatching, and data persistence
- Led UI/UX design in Figma in collaboration with an intern designer, producing complete user flows and high-fidelity mockups; delivered ~90% MVP completion before redirecting focus to a new venture

Nerd United

Blockchain Engineer

Lehi, UT (Hybrid)

April 2022 – April 2023

- Developed blockchain applications in Go and Solidity on EVM-based protocols for cryptocurrency clients; deployed 8 L1 GETH-based Proof-of-Authority testnets and 4 production mainnets (*technology available on request*)
- Wrote Go packages and Bash scripts to automate network provisioning, smart contract deployment, wallet funding, and Docker-based contract testing
- Built and deployed a production generative-AI pipeline in Python — orchestrating the OpenAI image-generation API, IPFS pinning, ERC-721 contract deployment to Sepolia Testnet, and end-to-end NFT minting as an automated multi-step workflow

- Validated smart contract behavior using the Brownie framework; implemented security controls and access enforcement across blockchain infrastructure

Skullcandy

Firmware Architect

Park City, UT

October 2018 – March 2022

- Architected firmware platform strategy across a portfolio of consumer audio products, establishing design patterns and standards that improved code quality and reduced per-product bring-up time
- Directed 3–5 firmware engineers through the full product lifecycle from prototype through design validation, production validation, and mass-production release; owned Apple MFi certification compliance and served as primary liaison to Apple's MFi program office
- Defined Bluetooth API contracts with third-party iOS/Android partners and implemented the firmware command interface for headphone feature control (EQ, battery reporting, device configuration)
- Authored a C# desktop application for USB firmware updates and a Python CLI build-environment toolchain with automated regression tests, eliminating manual flashing and compressing nightly validation cycles

Code Corporation

Senior Firmware Engineer

Draper, UT

August 2013 – October 2018

- Led 3–4 firmware engineers building advanced barcode reading systems from board bring-up through production; drove Agile/Scrum adoption within the team
- Reduced device firmware upgrade time from ~3 min to ~60 sec via USB Mass Storage optimization — a 3× improvement in field servicing throughput
- Implemented a secure bootloader, integrated a JavaScript engine for runtime-configurable device behavior, and wrote BLE firmware for the Nordic nRF52832 SoC
- Contributed Linux kernel patches for image sensor, GPIO/PWM, and USB Mass Storage support; implemented Apple MFi chip authentication, iAP2 protocol stack, and bare-metal peripheral drivers (GPIO, PWM, vibration, LCD)

TECHNICAL SKILLS

Languages	Python, TypeScript, Go, C, C++, C#, Objective-C, Swift, Kotlin
AI / LLM	Claude API (Anthropic SDK), multi-agent pipelines, LLM-output evals / backtesting, structured outputs, semantic search (embeddings), guardrails, model tiering & cost optimization
Web / Mobile	Next.js, React, Flutter (MVVM), Vite, GraphQL, Figma
Cloud / Backend	AWS Amplify, Lambda, DynamoDB, AppSync, API Gateway, Cognito, S3, CloudFront, Supabase
Embedded / HW	BLE/Bluetooth, Apple MFi/iAP2, RTOS, SPI/I2C/UART, USB (UDC, Mass Storage, HID), bare-metal ARM
Blockchain	EVM, Solidity, Go-Ethereum (GETH), IPFS, ERC-721, Brownie
Dev Tools	Git, Docker, Bash/Shell, REST/GraphQL APIs, Atlassian Suite

PROJECTS

Sentics — Multi-Agent Crypto Signal Pipeline

Python, Claude API, Next.js, Supabase, Vercel

Serverless multi-agent pipeline that screens top crypto assets: a deterministic technical-analysis stage sets directional calls, a Claude (Sonnet) synthesis stage writes rationale under a strict output whitelist so it can never override the ground-truth classification, and a Claude (Haiku) stage distills news into structured sentiment/catalyst reads. Model-tiered for cost (88% run-cost reduction), self-chaining batches keep each invocation under the serverless time limit, and a leak-free forward-tracking eval harness scores live directional calls against realized returns (edge, hit-rate, Sharpe, t-stat). github.com/jsanders5/sentics

Resumi — LLM Resume/Job-Matching Platform

Next.js, TypeScript, Claude API, Voyage embeddings

Full-stack app that ingests postings from multiple job boards, scores each against a candidate's resume via the Claude API (structured JSON breakdowns across skills, experience, domain, and requirements), and ranks matches with semantic search over Voyage embeddings. Prompt caching, exponential-backoff retries, and Sentry observability; deployed serverless on Vercel. github.com/jsanders5/resumi

PySando — Interactive Python Learning Platform

FastAPI, React, Claude API, pytest

Full-stack tutorial that runs learner code server-side in a resource-bounded sandbox and grades it against hidden pytest suites, paired with a Claude-powered tutor that grounds every reply in the live lesson, exercise, and the learner's current code and is

guardrailed never to reveal solutions. Async Claude integration with multi-turn conversation state, provider abstraction, and graceful degradation.

github.com/jsanders5/PySando

EDUCATION

University of Utah

M.S., Electrical & Computer Engineering

B.S., Computer Engineering

Salt Lake City, UT

2013

2013